Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 03

Subject: BSC102/AP **Date: 11/10/2026**
Max Marks: 10 **Deadline: 17/10/2026**

CO3: Determine the wavelength of light and refractive index of liquid using the interference phenomenon

Q. No.	Questions	Marks
1	Find the thickness of the soap film which appears yellow (wavelength 5896 Å) in reflection when it is illuminated by white light at an angle of 45°. Given refractive index of the film is 1.33.	2
2	A soap film 4×10^{-5} cm thick is viewed at an angle of 35° to normal. Calculate the wavelength of light in the visible spectrum that will be absent from the reflected light ($\mu = 1.33$).	2
3	In Newton's Ring experiment, the wavelength of light incident is 5×10^{-5} cm. If the diameter of the 10th dark ring is 0.5 cm, calculate the radius of curvature R of the given plano-convex lens.	2
4	In Newton's ring experiment, the diameter of the 15th ring was found to be 0.590 cm and that of the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, compute the wavelength of light used.	2
5	In a Newton's rings experiment, the diameters of the 10th and 20th dark rings are found to be 3.20 mm and 4.50 mm, respectively, when air is present between the lens and glass plate. When a transparent liquid is introduced, the diameters of the same 10th and 20th dark rings become 2.74 mm and 3.85 mm, respectively. Determine the refractive index of the transparent liquid using Newton's rings method	2